PLSC 40601

Week 2: Sample-splitting, shrinkage, cross-validation.

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Housekeeping

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Sample splitting

What is our goal in fitting a model?

- Given some data $(Y_1, \boldsymbol{X}_1), \ldots, (Y_N, \boldsymbol{X}_N)$, we fit a model, $\hat{f}(X)$.

- Suppose our goal is prediction for the next observation.

- Given X_{N+1}, we want to minimize

$$L\left(Y_{N+1}, \hat{f}(X_{N+1})\right) = \left(Y_{N+1} - \hat{f}(X_{N+1})\right)^2$$

- We may be interested not just in how our method performs on one specific observation, but how it performs in expectation

$$\mathrm{Err} = \mathrm{E}\left[L(Y, \hat{f}(X))\right]$$

- What should the expectation be taken over? Can/should we hold the data we used for fitting the model fixed?

What is our goal in fitting a model?

- Difference between conditional error and expected test error
  - Conditional test error:

$$\mathrm{Err}_{\mathcal{T}} = \mathrm{E}\left[L(Y, \hat{f}(X))|\mathcal{T}\right]$$

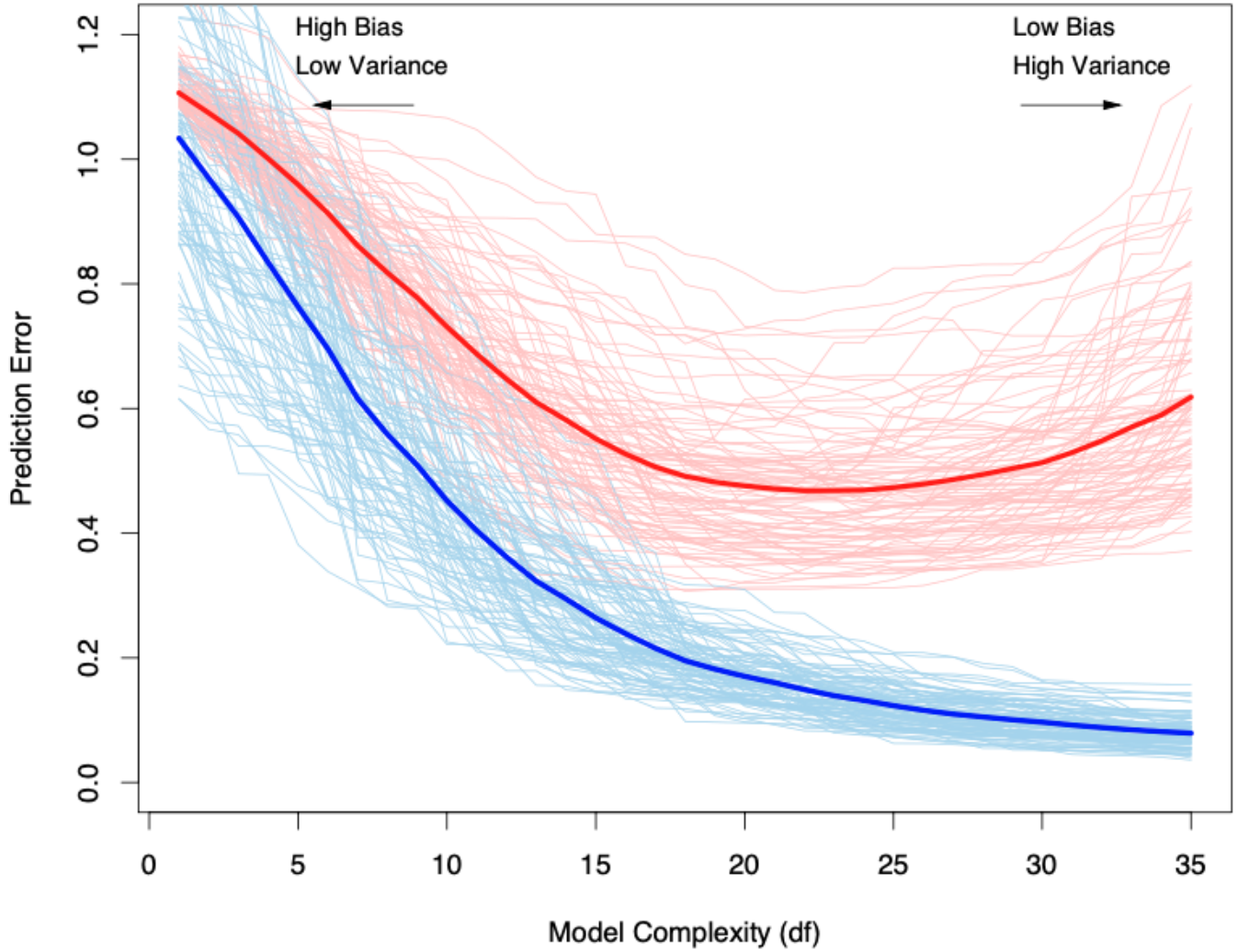
    Training set $\mathcal{T}$ is fixed.

  - Expected test error:

$$\mathrm{Err} = \mathrm{E}\left[L(Y, \hat{f}(X))\right] = \mathrm{E}\left[\mathrm{E}\left[L(Y, \hat{f}(X))|\mathcal{T}\right]\right]$$

- We may care about $\mathrm{Err}_{\mathcal{T}}$ for the model fit on our realized sample, but most practical methods estimate Err, the average test error over repeated samples.

What is our goal in fitting a model?



Hastie et al. (2009)

Blue is in-sample error, red is out-of-sample error.

What is our goal in fitting a model?

- We may want to use expected test error to **select among models**, or versions of models.

- And, once we have selected a version of a model, we may want to **assess** how a selected model performs.

Shrinkage.

Why shrinkage?

- With many covariates, interactions, or basis expansions, least squares can chase noise.
- Shrinkage deliberately adds some bias to reduce variance and improve out-of-sample prediction.
- This is especially attractive when many predictors are weak, or strongly correlated.

Ridge and lasso

$$\hat{\beta}(\lambda) = \arg\min_{\beta_0, \beta} \sum_{i=1}^{N} \left(y_i - \beta_0 - x_i^T \beta \right)^2 + \lambda P(\beta)$$

- Ridge: $P(\beta) = \sum_{j=1}^{p} \beta_j^2$.
- Lasso: $P(\beta) = \sum_{j=1}^{p} |\beta_j|$.
- Larger λ means more shrinkage. Ridge keeps all coefficients; lasso can set some exactly to zero.

Hastie et al. (2009)

Why lasso for adjustment?

- In the Bloniarz reading, treatment assignment is already randomized.
- Lasso is not identifying the treatment effect.
- It is used to improve precision by shrinking or selecting among many candidate adjustment variables.
- So the tuning target is predictive performance of the adjustment model.

What is our goal in fitting a model?

- We can't measure expected test error directly.

What is our goal in fitting a model?

- A procedure that allows us to estimate it:
  - Split data into three parts

  - Fit candidate models, or candidate penalty levels, to the training set.
  - Estimate prediction error of each candidate in the validation set.
  - Select the model or penalty with minimum validation error.
  - Then get generalization error of just that model on test set.

- Choosing λ is part of training, so the test set should be used only after tuning is finished. Still a version of winner's curse.

Cross-validation.

Cross-validation.

- We can potentially get more out of our data by cross-validating.

- For each candidate penalty λ, fit on one fold and validate on the held-out fold.

Version 1	Training	Validation
Version 2	Validation	Training

$$\widehat{\mathrm{Err}}_{CV}(\lambda) = \frac{1}{N}\sum_{i=1}^{N} L\left(y_i, \hat{f}_{\lambda}^{-k(i)}(x_i)\right)$$

$\hat{f}_{\lambda}^{-k(i)}$ is the fit from the training folds using penalty λ.

$$\hat{\lambda} = \arg\min_{\lambda} \widehat{\mathrm{Err}}_{CV}(\lambda)$$

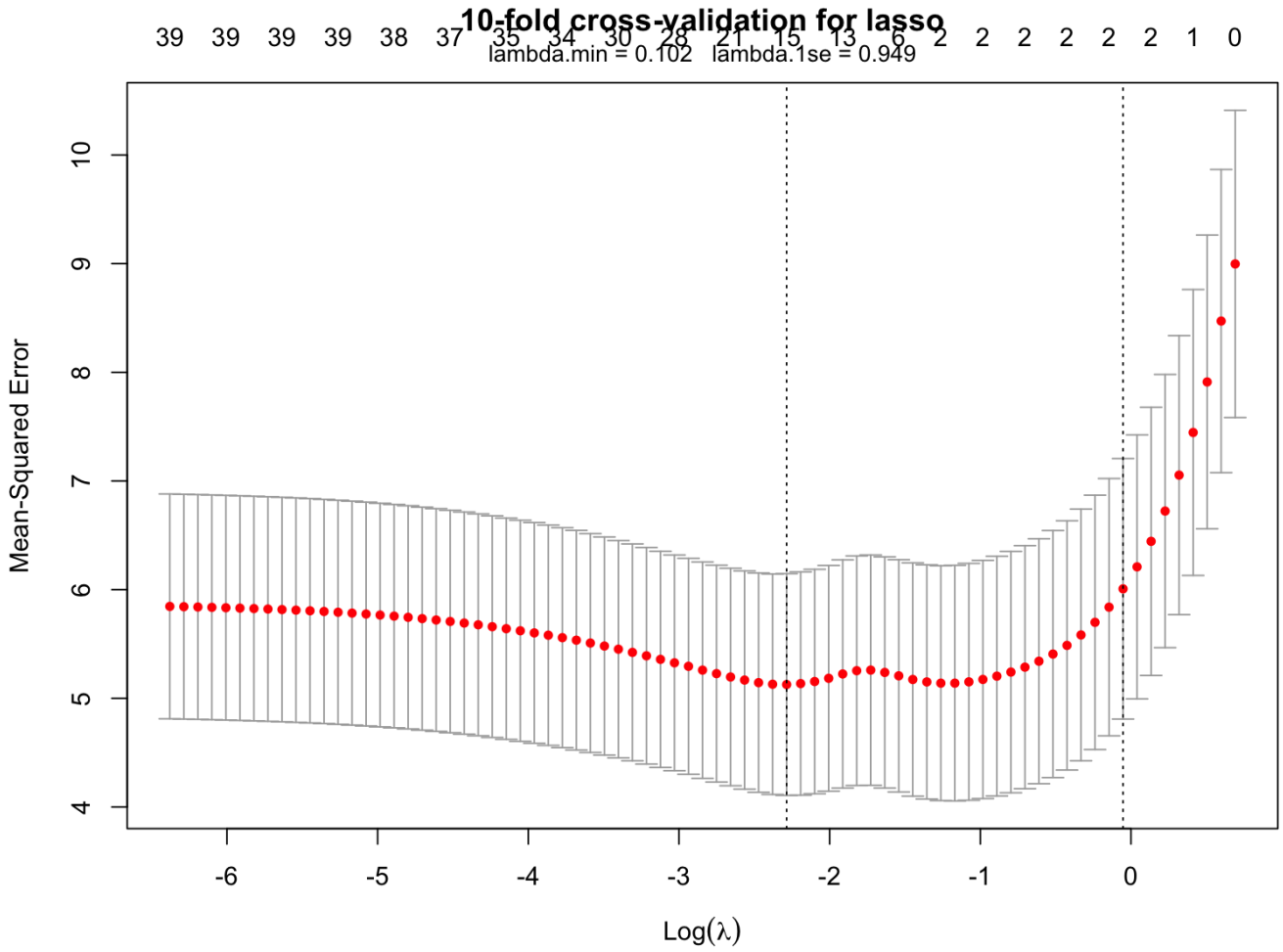
K-fold cross validation.

- Repeat the train/validate cycle across all K folds, average validation loss for each candidate λ.

Version 1	Training	Training	Training	Training	Validation
Version 2	Training	Training	Training	Validation	Training
Version 3	Training	Training	Validation	Training	Training
Version 4	Training	Validation	Training	Training	Training
Version 5	Validation	Training	Training	Training	Training

$$\widehat{\mathrm{Err}}_{CV}(\lambda) = \frac{1}{N}\sum_{i=1}^{N} L\left(y_i, \hat{f}_{\lambda}^{-k(i)}(x_i)\right)$$

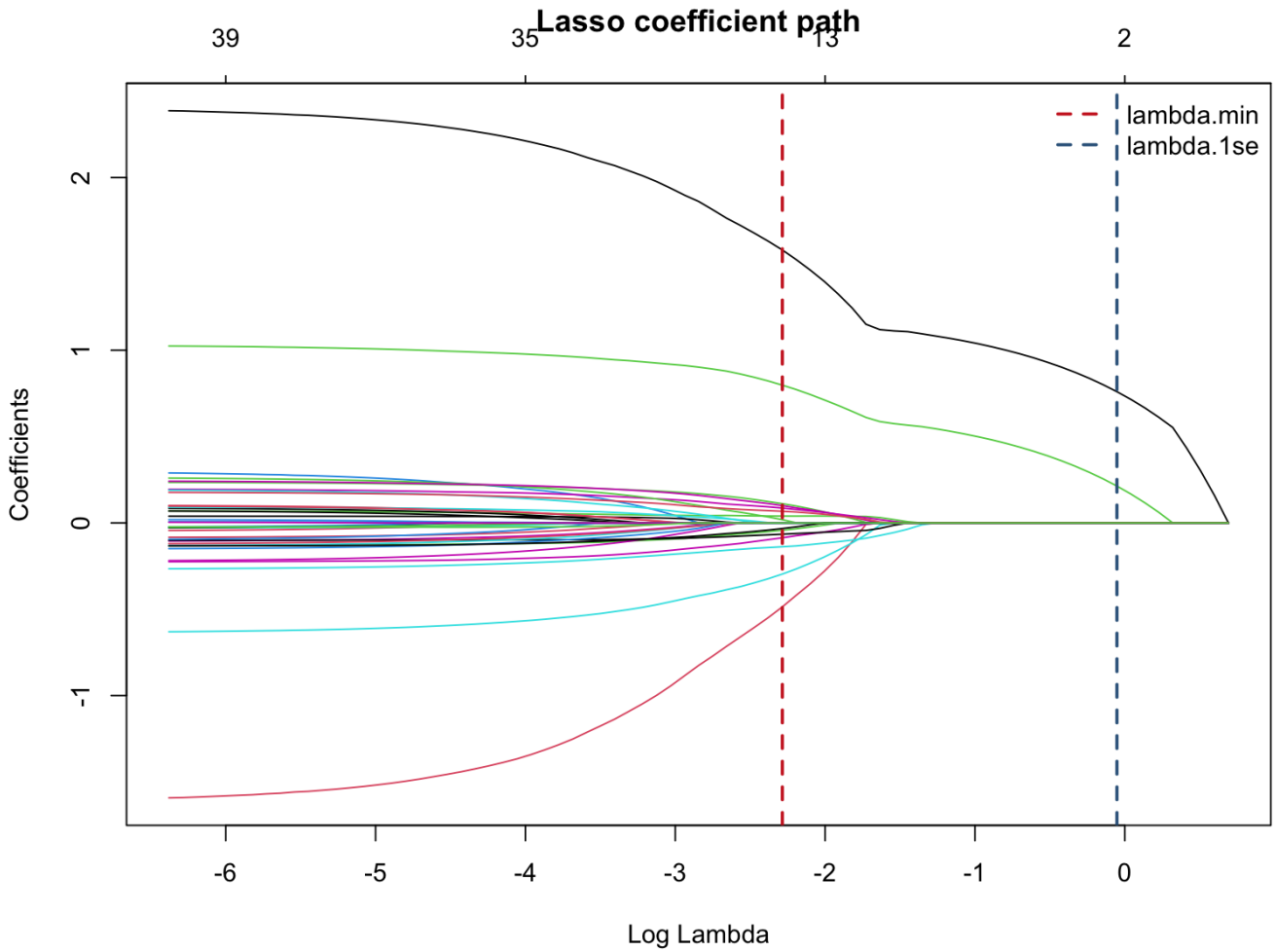
Cross-validation for lasso



- `cv.glmnet` reports `lambda.min` and `lambda.1se`; The one-standard-error rule picks the simplest model whose estimated CV error is within one standard error of the minimum estimated CV error.

$$\hat{\lambda}_{1se} = \max\left\{\lambda : \widehat{\mathrm{Err}}_{CV}(\lambda) \leq \widehat{\mathrm{Err}}_{CV}(\hat{\lambda}_{\min}) + \widehat{\mathrm{SE}}\left(\widehat{\mathrm{Err}}_{CV}(\hat{\lambda}_{\min})\right)\right\}$$

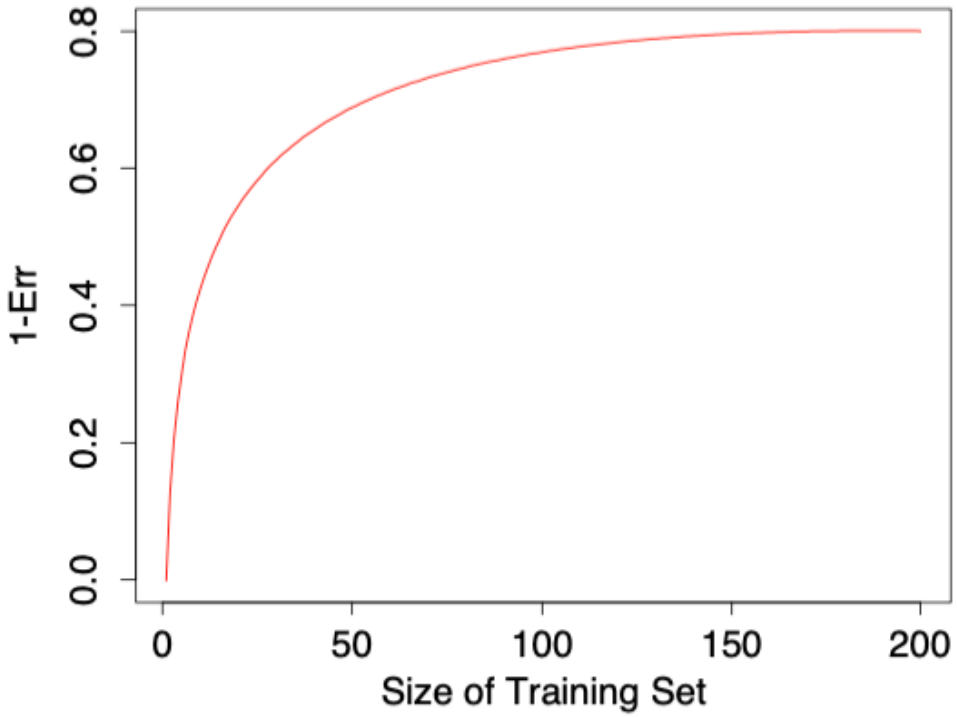
Lasso coefficient path



- As λ falls, more coefficients enter the model.
- Cross-validation chooses a point along this path.

Cross-validation.

- How do we pick K?
- $K = N$? Low bias, possibly high variance (our prediction sets are very similar).
- $K = 5$? Lower variance, possibly higher bias. How much does the prediction change as we change the size of the data set?



Hastie et al. (2009)

Cross-validation.

- Rule of thumb is often 5 or 10 folds.
- For lasso, the other practical choice is whether to report `lambda.min` or the more conservative `lambda.1se`.

References I

Hastie, T., Tibshirani, R., and Friedman, J. H. (2009). *The elements of statistical learning: data mining, inference, and prediction*, volume 2. Springer.